

Lecture 4

September 3

least squares...

Learning objectives

- The least squares formulation and how it produces equations for estimating unknown quantities.
- The differences among Mayer's, Boscovich's, Laplace's, and Legendre's principles for defining a good rule for combining observations.
- How measurement error, differences between errors, rounding, and scientific judgment shaped the measurement of the meter and Legendre's analysis.

What did Boscorich do?

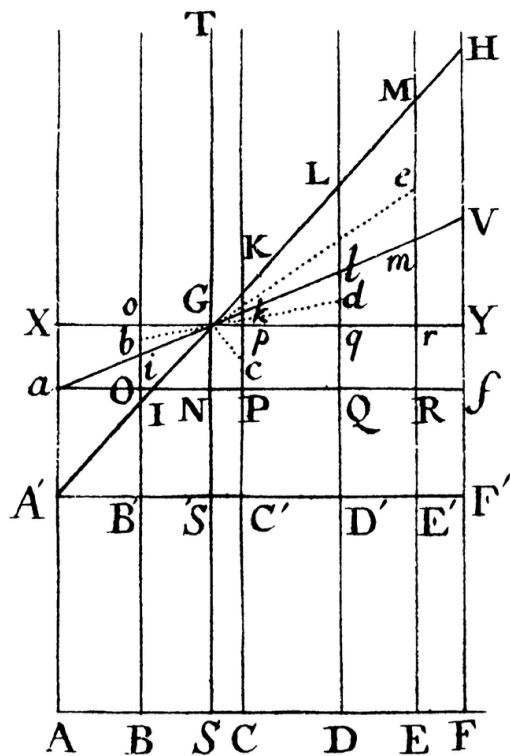


Figure 1.4. Boscovich's algorithm. The horizontal axis AF gives $\sin^2 \theta$, where θ is the latitude of the midpoint of the arc; the vertical axis AX gives the arc length in toises per degree. The five arcs are indicated by a, b, c, d, and e; G is the center of gravity. (From Boscovich and Maire, 1770, plate I.)

Geometric solution for
minimize $\sum_i |a_i - z - y \sinh^2 \theta_i|$
such that $\sum_i (a_i - z - y \sinh^2 \theta_i) = 0$

a_0 is an arc length of a
1° of latitude at latitude θ_i
given in fms/degree

let w_i be the arc length in degrees.

Real measurement error in a_i or in $a_i w_i$?

Laplace, again.

First, that the sum of the errors committed in the measures of the whole arcs, ought to be zero. Second, that the sum of all these errors, taken positively, ought to be a *minimum*. By considering, in this manner, the whole arcs, instead of the degrees which have been deduced from them, we shall give to each of these degrees so much more influence, in the computation of the ellipticity of the earth, as the corresponding arc is of greater extent, which ought to be the case. (Laplace, 1799–1805, vol. 2, p. 134; Bowditch, 1832, pp. 434–437)

Laplace's minimization

$$\min \sum |a_i w_i - z v_i - y w_i \sin^2 \theta_i| \quad \text{such that } \sum (a_i w_i - z v_i - y w_i \sin^2 \theta_i) = 0$$

$$\min \sum \underbrace{w_i}_{\text{intrinsic value/quality of the observation}} |a_i - z - y \sin^2 \theta_i| \quad \dots$$

Adrien-Marie Legendre

1752-1833 all in Paris

- Published least squares method in 1805...

by 1809 Gauss claimed that he
created it!

- A bit of a subversive: refused to vote for the government candidate for the Academy and had his pension cut. (Interestingly the candidate he backed won... because Science).

Who is Legendre?

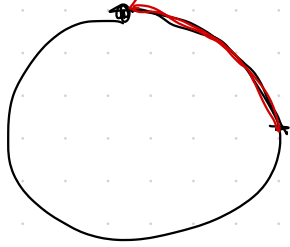


OR



What is a meter?

- $1/10^{10}$ of the meridian quadrant



Delambree goes north

Mechain goes south to Barcelona

(1) measurement in Paris Monjuic

(2) another measurement is different.

- Distance traveled by light in a vacuum in a very short amount of time.

Is lying about data bad?

- Mechain reports only the Montjuic measurement
- 1804 Mechain dies before correcting his measurements.
- Delambre finds the notations, decides not to correct the correct.

Actual measurement : 10002 cm instead of 10000.

yes

Legendre's principle

Of all the principles that can be proposed for this purpose, I think there is none more general, more exact, or easier to apply, than that which we have used in this work; it consists of making the sum of the squares of the errors a minimum. By this method, a kind of equilibrium is established among the errors which, since it prevents the extremes from dominating, is appropriate for revealing the state of the system which most nearly approaches the truth. (Legendre, 1805, pp. 72–73)

Legendre's method

$$E_i = a_i + b_i x + c_i y + f_i z + \dots$$

$$\text{goal } \min_{\text{ing}} (E_1^2 + E_2^2 + \dots) = \min \sum_i E_i^2$$

$$\min \left[(a_1 + b_1 x + c_1 y + f_1 z + \dots)^2 + (a_2 + b_2 x + c_2 y + f_2 z + \dots)^2 + \dots \right]$$

take derivative, set equal to 0

$$\frac{\partial}{\partial x} 0 = 2(a_1 + b_1 x + c_1 y + f_1 z + \dots)b_1 + 2(a_2 + b_2 x + c_2 y + f_2 z + \dots)b_2 + \dots$$

$$\text{combine like terms} = 2 \left(\sum_i a_i b_i + x \sum_i b_i^2 + y \sum_i b_i c_i + \dots \right)$$

$$\frac{\partial}{\partial y} 0 = 2(a_1 + b_1 x + c_1 y + \dots)c_1 + 2(a_2 + b_2 x + c_2 y + \dots)c_2 + \dots$$

Same # of equations as unknowns.

Solve.

More compact notation (page 35)

'equations of condition'

$$0 = a_i + w + b_i x + c_i y + d_i z$$

(w, x, y, z) unknown

$$0 = \sum_i k_{ij} a_i + \sum_i k_{i1} w + \sum_i k_{i2} b_i x + \sum_i k_{i3} c_i y + \sum_i k_{i4} d_i z$$

Mayer's method

$$k_{i1} = 1 \text{ if } b_i \leq \underline{b}, 0 \text{ otherwise}$$

$$k_{i2} = 1 \text{ if } b_i \geq \overline{b}, 0 \text{ otherwise}$$

$$k_{i3} = 1 \text{ if } \underline{b} < b_i < \overline{b}, 0 \text{ otherwise}$$

Laplace

$$k_{i1} = 1$$

$$k_{i2} = -1 \text{ if } b_i < B, +1 \text{ if } b_i > B$$

$$k_{i3}, k_{i4} \dots$$

Mayer, Laplace update to Mayer, Boscorich?

[Boscorich can't easily be represented like this
because the combination rule depends on the
median residual.

Legend

$$K_{i1} = 1$$

$$K_{i2} = b_i$$

$$K_{i3} = c_i$$

$$K_{i4} = d_i$$

One more (modern) way to write least squares

$$E_i = a_i - (w + b_i x + c_i y + d_i z)$$

Error or difference in error?

$$E' - E'' = 0.002923 + \epsilon(2.192) - \alpha(0.563)$$

$$E'' - E''' = 0.003100 + \epsilon(2.672) - \alpha(0.351)$$

$$E''' - E^{iv} = -0.001096 + \epsilon(2.962) + \alpha(0.047)$$

$$E^{iv} - E^v = -0.001808 + \epsilon(1.851) + \alpha(0.263)$$

Comme il importe de considérer les erreurs séparément, on regardera comme une nouvelle inconnue l'erreur E'' , par exemple, et on aura les cinq équations :

$$\begin{aligned} E' &= E'' + 0.006023 + \epsilon(4.864) - \alpha(0.914) \\ E'' &= E''' + 0.003100 + \epsilon(2.672) - \alpha(0.351) \\ E''' &= E^{iv} \\ E^{iv} &= E'' + 0.001096 - \epsilon(2.962) - \alpha(0.047) \\ E^v &= E'' + 0.002904 - \epsilon(4.813) - \alpha(0.310) \end{aligned} \quad (b)$$

When do you round?

Legendre's own instruction, printed pp. 73–74

Mais on aura soin d'abrégé tous les calculs, tant des multiplications que de la résolution, en n'admettant dans chaque opération que le nombre de chiffres entiers ou décimaux que peut exiger le degré d'approximation dont la question est susceptible.

Translation. Care should be taken to abbreviate all the calculations, both the multiplications and the solution, admitting in each operation only the number of integer or decimal digits required by the degree of approximation of which the question is susceptible.

legendre's math

$$r_1 = g + 0.006023 + 4.864\varepsilon - 0.914\alpha,$$

$$r_2 = g + 0.003100 + 2.672\varepsilon - 0.351\alpha,$$

$$r_3 = g,$$

$$r_4 = g + 0.001096 - 2.962\varepsilon - 0.047\alpha,$$

$$r_5 = g + 0.002904 - 4.813\varepsilon - 0.310\alpha.$$

$$d = \begin{pmatrix} 0.006023 \\ 0.003100 \\ 0 \\ 0.001096 \\ 0.002904 \end{pmatrix}, \quad b = \begin{pmatrix} 4.864 \\ 2.672 \\ 0 \\ -2.962 \\ -4.813 \end{pmatrix}, \quad c = \begin{pmatrix} -0.914 \\ -0.351 \\ 0 \\ -0.047 \\ -0.310 \end{pmatrix}$$

$$r = g\mathbf{1} + d + b\varepsilon + c\alpha,$$

$$\begin{aligned} g &= -\frac{0.013123}{5} + \frac{0.239}{5}\varepsilon + \frac{1.622}{5}\alpha \\ &= -0.0026246 + 0.0478\varepsilon + 0.3244\alpha \end{aligned}$$

$$\begin{aligned} 0 &= 5\mathbf{E}'' + 0.013123 - \varepsilon(0.239) - \alpha(1.622) \\ \text{donc } \mathbf{E}'' &= -0.002625 + \varepsilon(0.048) + \alpha(0.324). \end{aligned}$$

1

Where should the shuttle stop?

When every additional mile of walking adds the same amount of inconvenience

Seven families live along one straight road. Their homes are at mile markers 1, 1, 2, 3, 4, 5, and 12. The town will place one shuttle stop somewhere on this road. Every family uses the shuttle equally often.

For this decision, inconvenience is simply walking distance. One family walking two miles creates exactly as much total inconvenience as two families walking one mile each. Ignore construction cost, traffic, and every other consideration.

2

What common length should the springs have?

When large displacements grow disproportionately costly

Seven springs have the same stiffness but different natural lengths: 1, 1, 2, 3, 4, 5, and 12 centimeters. They must be clamped between two rigid bars, so all seven will be forced to the same final length a . A spring may be stretched or compressed; only the size of the displacement matters.

For these springs, forcing a spring twice as far from its natural length makes the energy stored in that spring four times as large. The engineers want the common length that makes the total stored energy as small as possible.

- A displacement of 2 stores four times as much energy as a displacement of 1.
- A displacement of 4 stores four times as much energy as a displacement of 2.
- You do not need any additional spring physics; these comparisons contain all the information you need.

3

Which connector size should the workshop make?

When a near miss is just as unsuccessful as a large miss

Seven old instruments need replacement connector pins of sizes 1, 1, 2, 3, 4, 5, and 12 millimeters. The workshop has enough time and material to manufacture one connector size in bulk.

A connector works only when its size matches an instrument exactly. A connector that is 1 millimeter too small is just as useless as one that is 10 millimeters too small. There is no partial fit, and connectors cannot be modified afterward. Each instrument is equally important.

The workshop wants to make as many of the seven instruments operational as possible.

Next class
probability...